

Collection and Compilation of Water Pipeline Field Performance Data

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Abstract

The objective of this project is to collect, compile and analyze high quality field data on pipeline reliability performance for water pipelines of different materials. This project will be completed in two phases: Phase I will include the data collection from the United States Bureau of Reclamation (USBR) and their 17 Western States, as well as the U.S., Canadian, and Australian Water Utilities for different materials including cast iron, ductile iron, reinforced concrete, steel, pre-stressed concrete, PVC, AC, and others. Canadian and Australian utilities' data and knowledge will be stored as a subset of the national pipeline database. Phase I will also include the development of a Web-based, GIS-enabled water pipelines database to aggregate and standardize the collected data, and further to identify the missing data deemed critical to understanding water pipeline performance. These tasks will be geared to meet the Phase II objectives of collecting and/or generating the missing data, and the establishment of an understanding of the general state of buried water pipeline infrastructure, failure rates, general effectiveness of corrosion control measures, and the calculation of failure rates for each water pipeline material. Virginia Tech will protect data and database as per federal requirements under export-import control law. The proposed research has the following five objectives:

1. Developing a standardized data and metadata structure for verifying, storing, updating, retrieving, and exporting/importing water pipeline infrastructure attributes;
2. Creating a GIS-driven Web-based platform for developing, analyzing, validating, implementing, and benchmarking of models and tools for pipeline asset management;
3. Piloting of PIPEiD with utilities of various sizes across the US for demonstration;
4. Establishing a protocol for water pipeline data security, sanitization, and publication; and
5. Developing a plan to promote and sustain effort by engaging the water industry.

This paper presents the national effort for water pipeline data collection and compilation methodologies. This project will determine the state of the knowledge in water utilities, other industries, and large Internet of Things (IoT) firms to lay the groundwork to identify key issues that can be addressed by data collection and big data analytics for water pipeline infrastructure systems.

1. Data Standard to Estimate Likelihood of Failure

The life cycle of pipe systems consists of planning; engineering; manufacturing; construction; operation and maintenance; and repair, rehabilitation, and replacement. A data Standard is needed to support advanced pipeline infrastructure asset management. Performance management is defined as managing the pipeline infrastructure to minimize the total cost of owning and operating it while delivering service levels to meet or exceed customers’ expectations at a level of acceptable risk to the utility. Many factors affect the performance of water pipes (Figure 1). It is not fully understood how these factors affect pipeline performance deterioration.

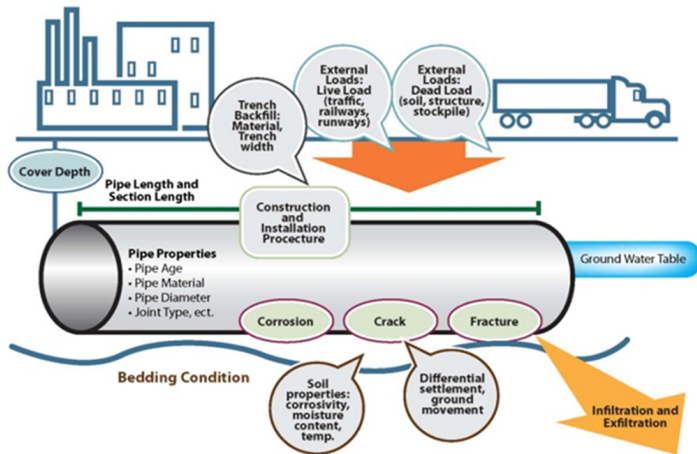


Figure 1. Some of the Factors Affecting Performance of Pipeline Infrastructure

Attributes	
OBJECTID	
SHAPE	
StructID	
Content	
Type	
Alias: Type	
Data type: String	
Width: 50	
Precision: 0	
Scale: 0	
Definition: type of pipe	
Material	
Alias: Material	
Data type: String	
Width: 50	
Precision: 0	
Scale: 0	
Definition: pipe material	

The foundation of a database is its Metadata Standard. Metadata is data about data; it is used to facilitate the understanding, characteristics, and management usage of data. This typically includes the name of the field, data type, width and definition. The Figure 2 illustrates the metadata format for water pipelines.

Figure 2. Metadata Standards for Water Pipelines

Below are brief definitions of the terms used.

- Alias: another name used for the field;
- Data type: the type of data acceptable in the field (i.e. string or number etc.);
- Width: the length of the field (i.e. number of characters or numbers); and
- Definition: the description of the attribute.

Metadata summarizes basic information about data, which can make finding and working with particular data easier. In addition to the metadata file, domains and ranges have been defined for each attribute to prevent the addition of false data by mistake or by intention (see Figure 3). The domain list is the summary of all the

values used by the utilities and the possible values noted by researchers. However, the domain list is not exhaustive, but is one that needs to be updated frequently. The metadata file can also be updated and changed according to the needs of the utilities.

Attributes of Conduits										
OBJECTID	SHAPE	Structure ID	Content	Type	Material	FromNode	ToNode	XS shape	Diameter	From Invert Elevation
1	Polyline	215-318 215-321	C	gravity	RCP	215-318	215-321	circular	30 30	293.6
2	Polyline	215-317 215-318	C	gravity	RCP	215-317	215-318	circular	30 30	297.6
3	Polyline	215-322 215-321	S	gravity	Brick	215-322	215-321	circular	8 8	306.5
4	Polyline	215-321 220-028	C	gravity	Brick	215-321	220-028	circular	30 30	293.6
5	Polyline	219-060 219-059	S	gravity	Centrifugally Cast P	219-060	219-059	circular	8 8	449.2
6	Polyline	219-442 219-076	S	gravity	cast iron	219-442	219-076	circular	8 8	9.7
7	Polyline	219-153 219-154	S	gravity	Cast In Place Pipe	219-153	219-154	circular	8 8	481.1
8	Polyline	219-413 219-060	S	gravity	Corrugated Metal Pi	219-413	219-060	circular	8 8	9.7
9	Polyline	219-155 219-156	S	gravity	concrete	219-155	219-156	circular	8 8	456.8
10	Polyline	219-449 219-159	S	gravity	copper	219-449	219-159	circular	8 8	9.7
11	Polyline	220-341 220-008	S	gravity	ductile iron	220-341	220-008	circular	8 8	0
12	Polyline	220-005 220-004	S	gravity	Fiber Reinforced Pla	220-005	220-004	circular	8 8	443.8
13	Polyline	220-323 220-322	C	gravity	High Density Pipe	220-323	220-322	circular	8 8	439.4
14	Polyline	220-322 220-321	C	gravity	polyethylene	220-322	220-321	circular	8 8	439.4
15	Polyline	220-350 220-063	S	gravity	COON	220-350	220-063	circular	8 8	0
16	Polyline	220-353 220-075	S	gravity	COON	220-353	220-075	circular	8 8	0
17	Polyline	220-088 220-089	S	gravity	COON	220-088	220-089	circular	8 8	389.3
18	Polyline	220-347 220-056	S	gravity	COON	220-347	220-056	circular	8 8	0
19	Polyline	220-076 220-319	S	gravity	COON	220-076	220-319	circular	8 8	402.2
20	Polyline	220-106 220-105	S	gravity	COON	220-106	220-105	circular	8 8	357.5
21	Polyline	220-024 220-023	S	gravity	COON	220-024	220-023	circular	8 8	336.9
22	Polyline	220-097 220-098	S	gravity	COON	220-097	220-098	circular	8 8	392.7
23	Polyline	220-357 220-098	S	gravity	COON	220-357	220-098	circular	8 8	0
24	Polyline	220-099 220-098	C	gravity	COON	220-099	220-098	circular	8 8	388.2
25	Polyline	220-098 220-100	C	gravity	COON	220-098	220-100	circular	8 8	384.7
26	Polyline	220-108 220-109	C	gravity	CP	220-108	220-109	circular	12 12	316.3
27	Polyline	220-082 220-080	S	gravity	COON	220-082	220-080	circular	8 8	387.6
28	Polyline	220-025 220-024	S	gravity	COON	220-025	220-024	circular	8 8	337.1
29	Polyline	0220-081 0220-080	D	gravity	COON	0220-081	0220-080	circular	8 8	429.2
30	Polyline	220-113 220-114	C	gravity	RCP	220-113	220-114	circular	30 30	257.6
31	Polyline	220-032 220-111	C	gravity	RCP	220-032	220-111	circular	30 30	276.6
32	Polyline	220-031 220-032	C	gravity	RCP	220-031	220-032	circular	30 30	281.8
33	Polyline	220-030 220-031	C	gravity	RCP	220-030	220-031	circular	30 30	282.3
34	Polyline	220-029 220-030	C	gravity	RCP	220-029	220-030	circular	30 30	281.3
35	Polyline	220-114 220-172	C	gravity	RCP	220-114	220-172	circular	30 30	262.6
36	Polyline	220-115 220-172	C	gravity	RCP	220-115	220-172	circular	30 30	262.6

Figure 3. Adding Values to the Attribute Table of Conduits for QA/QC

2. Data Standard for Failure and Forensic Analysis

The failure modes of water pipes are complex. The rate of deterioration is affected by a number of factors. Although there are many condition assessment technologies and accelerated testing methods available, these techniques do not provide all of the required data on the factors affecting deterioration. Earlier efforts have been made to capture data on pipe failure; this data has been used to develop some of the existing deterioration models. However, failure data alone is limited in terms of understanding pipeline deterioration. Forensic analyses of pipe samples extracted while in service, as well as after failure, are crucial in acquiring a comprehensive understanding of the rate of deterioration over time. Utilities need feasible and sound protocols to analyze deteriorated pipes. Failure and forensic analysis provides better understanding of pipe failure modes and mechanisms for performance management, as shown in Figure 4.



Figure 4. Field Failure and Forensic Analysis of Water Pipeline Infrastructure

Experiments to simulate actual field scenario in laboratory environments have been conducted by several research facilities, but most have failed to simulate the actual field conditions. It has been reported that the stress distributions generated within the soil mass and at the pipe surface in deep pipe burial are very different compared to the actual field scenario. Virginia Tech has recognized this issue and has proposed to construct a new *Smart Pipe Research Facility* to provide world-class capability for pipeline systems research. In the meantime, water utilities have developed various protocols for failure and forensic analysis. Figure 5 is an example of a data sheet for field data collection of a pipe sample during repair.

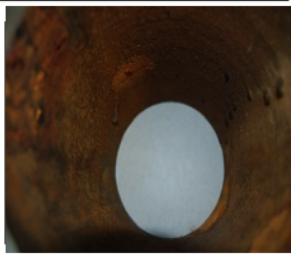
WESTERN VIRGINIA WATER AUTHORITY		Pipe Sample	
ID	SAMPLE-1329509487	Original Repair ID	REPTST
Pipe ID	PIPE-000012	Location	Address
Team leader	Asphalt Solutions	Date Sample Obtained in Field	Friday, February 17, 2012 8:10 PM
Start length (ft)	10	Sample length (ft)	10
Diameter (in)	8	Pipe material	Cast Iron
Dissimilar Materials Present	☒ Yes ☐ No	Dissimilar Material Adjacent to Pipe Material	Galvanised Iron
5lb Soil Sample Collected	☒ Yes ☐ No	Cathodic protection	☒ Yes ☐ No
Surface material	Sand or Gravel	Surface type	Road
Depth of cover (ft)	8	Depth of ground water table from surface (ft)	4
Joint type	Flanged	Bedding	None
Failure location	Pipe	Failure position	Invert
Failure type	Circular Fracture	Date/Time Sample Delivered to VT	Friday, February 17, 2012 8:30 PM
Manufacture era	☐ Pre 1930 ☐ 1930-1950 ☒ 1950-1970 ☐ Post 1970	Notes	Test Notes
Photo Panel			
External photo		Internal photo	
Data from WVWA Office			
Pipe Age	60	Design Life	75
Breakage history	3	Water pH	6.5
Pressure (psi)	78	Pipe Manufacturer	Griffin Pipe

Figure 5. Field Data and Pipe Sample Collection for Failure Analysis

Virginia Tech has established a protocol for collection of aged pipe samples from various water utilities and performance of laboratory experiments to better understand the deterioration behavior. Figure 6 represents water pipe sample sheet for laboratory testing and data collection.

Tensile Test Results from VT Laboratory	
Ultimate tensile stress	
Young's modulus	
Secant modulus	
Stress-strain relationship	
Initial tangent modulus	

Ring Bearing Test Results from VT Laboratory	
Tensile strength	
Rupture modulus	

Soil Results from VT Laboratory	
Soil type	Clay
Sulphide concentration	
Soil pH	
Drainage / Moisture content	
Soil resistivity	

Geometric Results from VT Laboratory	
External diameter (in)	8.6
External protection	Bitumen/Tar
External protection status	Most of the surface
Tuberculation	Some of the surface
Tuberculation removal	Easy
Min wall thickness (in)	0.1
Graphitisation	Crazed
Graphitisation/Pits	Some External
Max pit depth (in)	0.3
Max ext pit depth (Cut edge) (in)	0.2
Encrustation	Most of the surface
Internal protection	Epoxy Resin
Internal protection status	Whole surface
Max tuberculation height (in)	0.8
Pitting	Most of the surface
Max wall thickness (in)	0.4
Graphitisation status	Easily Detached
Remaining clear bore (in)	7.5
Casting defects	Uneven Casting
Max int pit depth (Cut edge) (in)	0.3

Figure 6. Field Data and Pipe Sample Collection for Failure Analysis

3. Data Standard to Represent Consequence of Failure

Risk is defined as: “The chance of something happening that will impact upon objectives, and is measured in terms of a combination of the likelihood and consequences of events.” Risk is often specified in terms of likelihood of failure and consequence of failure. The consequence of failure determines the impact due to asset failure. The consequence of a failure may lead to loss and the loss can be quantified into direct and indirect costs, environmental costs, and societal costs.



Figure 7. Consequence of Water Pipeline Infrastructure System Failure

Consequences of events can be evaluated from real world observations, tests, models or expert opinions (see Figure 7). In a qualitative risk assessment using the matrix method, scales are used to describe the magnitude of the consequence. An example of possible qualitative scale to rate the consequence of different risks occurring is shown in Figure 8. Risk levels are then displayed in a matrix that applies rules to combine likelihood and consequence into a level of risk.

		Criteria						
	Rating	Social	Environment	Economic	Regulation	Reputation	Renewal Complexity	Pipeline Scenario
A	Extreme	A1	A2	A3	A4	A5	A6	A7
B	Major	B1	B2	C3	B4	B5	B6	B7
C	Moderate	C1	C2	D3	C4	C5	C6	C7
D	Minor	D1	D2	E3	D4	D5	D6	D7
E	Negligible	E1	E2	E3	E4	E5	E6	E7

Figure 8. Consequence Rating Matrix for Water Pipeline Asset Management

4. Data Standard to Conduct Financial and Economic Life-Cycle Analysis

Responsible economic decision making for infrastructure investment must include an analysis of all costs involved in the life of the asset (Figure 9) and risk analysis (Figure 10). These include:

- Capital (Outlay or Depreciation) and Opportunity Cost of Capital
- Maintenance Costs, Operating Costs, and Administrative Costs
- Renewal Costs (Repair, Rehabilitation, and Replacement)
- Risk Cost and Total Business Impact Cost

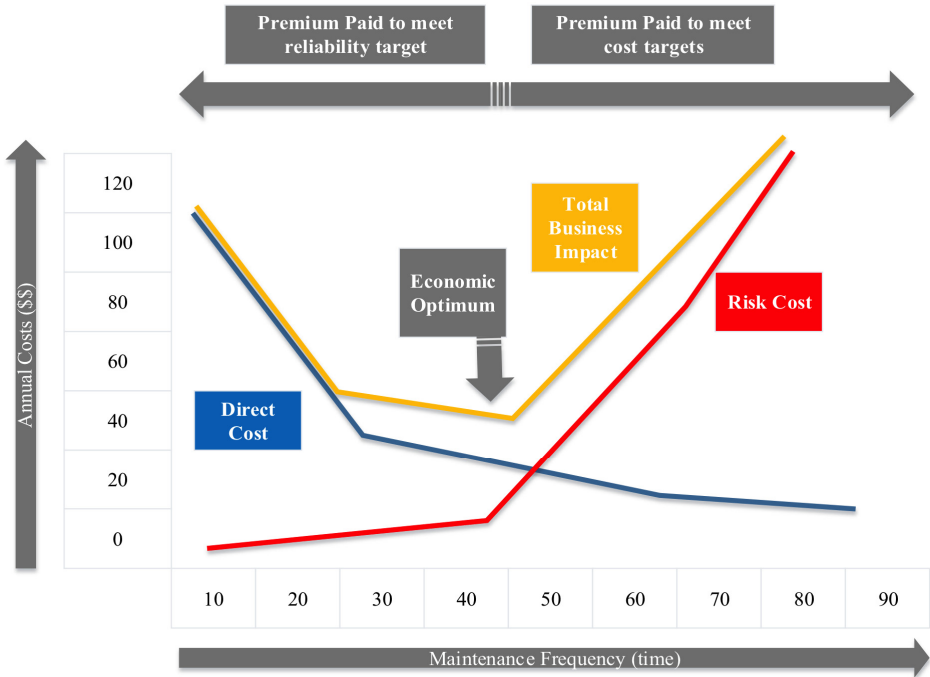


Figure 9. Risk Management based on Maintenance Frequency

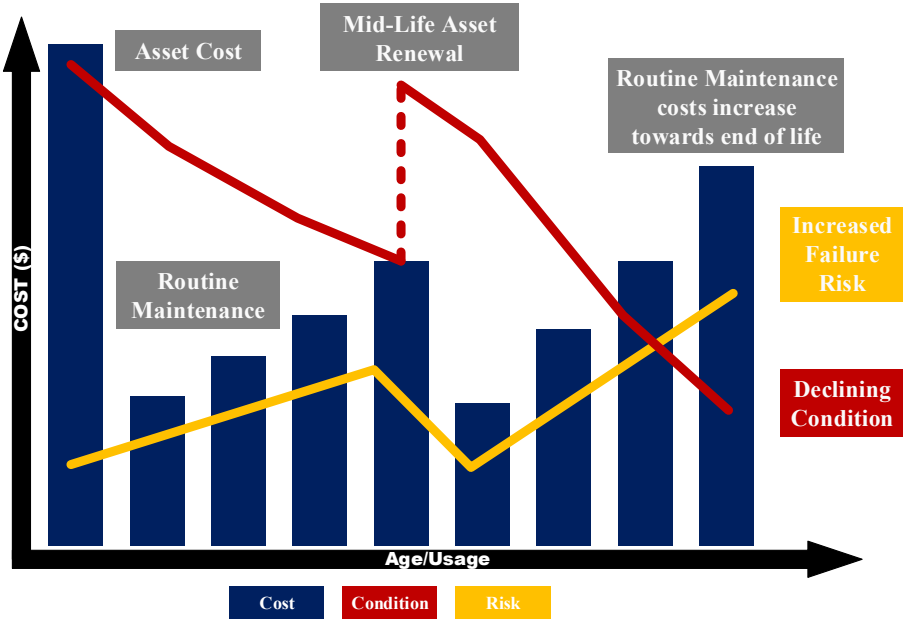


Figure 10. Risk Management based on Declining Asset Condition

5. Data Standard to Capture Performance, Sustainability, and Resiliency

Asset Management is the major driver in the water industry for risk management. The deterioration of the infrastructure has resulted in severe budget ramifications and has impacted the level of service for the water, wastewater and stormwater utilities. Historically, the asset management plans for utilities have focused on the performance aspects of the assets. Performance parameters have been collected and analysis of the data has led to the development of a performance index from 1 to 5 for the utility. Recently, utilities have begun discussing the sustainability and resiliency of the assets. In the original performance parameters there were some that related to sustainability and resiliency. It is becoming apparent that it is imperative to rate a utility with an index for performance, sustainability and resiliency. The following definitions are presented to establish the standard term for industry.

- **Performance Management:** managing the pipeline infrastructure to minimize the total cost of owning and operating while delivering service at a level of acceptable risk to the utility.
- **Sustainability Management:** considers the ability of the pipeline systems to operate at the pre-defined levels of service (Social, Economic, and Environmental) for indefinite time.
- **Resilience Management:** allows the pipeline infrastructure systems to recover from disruption events (Natural, Manmade, Aging) within acceptable time and cost limits.

The asset management framework incorporates basic elements that build on and complement one another to provide sustainable municipal infrastructure asset management. Unlike other asset management structures, the new framework links standard asset management concepts, information systems, and sustainable and resilience management practices. Ideally, this framework provides utilities with a support system that handles short and long term holistic asset management planning (Sinha & Eslambolchi, 2006). The framework is outlined in Figure 11.

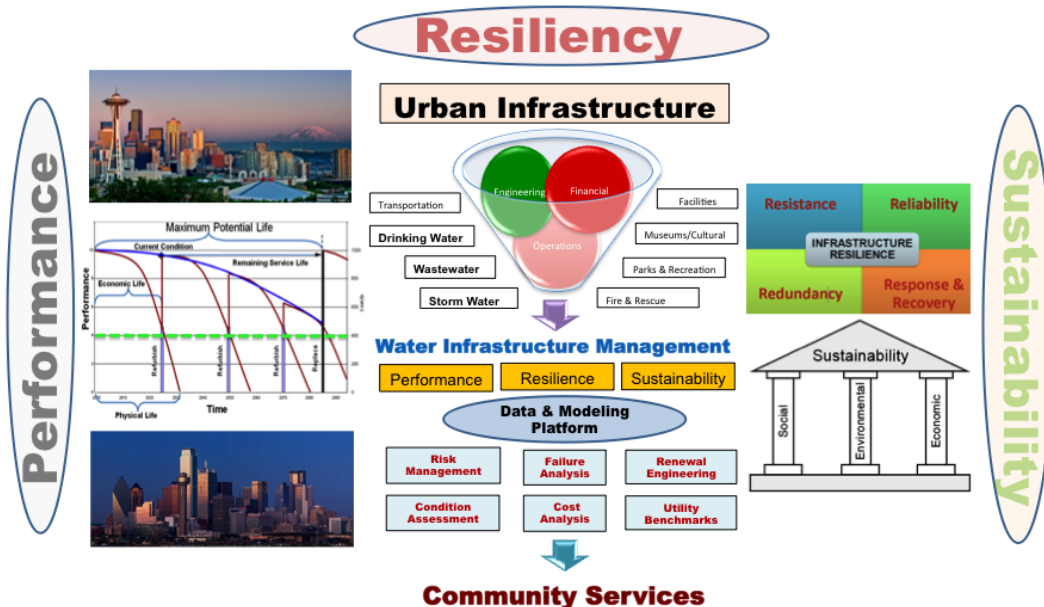


Figure 11. Pipeline Performance, Sustainability, and Resiliency Management

In most cases the asset management models have incorporated condition and financial issues. In the US, there has been focus on condition assessment, deterioration curves and triple bottom line analysis. In the past five years, EPA and other agencies have been discussing sustainability and resiliency. Each of these areas has received attention on an individual basis. There are parameters and interdependencies that cross all three areas of performance, sustainability and resiliency. Figure 12 presents the research on development of the governing parameters associated with each of three areas (Performance, Sustainability, and Resiliency).

PERFORMANCE	SUSTAINABILITY	RESILIENCY
• Structural Condition • Internal Environment • External Environment	• Societal Issues • Environmental Factors • Economical Factors	• Assessment Strategy • Mitigation Plan • Recovery Action

Figure 12. Performance Management based on Sustainability and Resiliency

The holistic asset management framework incorporates various aspects of asset management such as data collection, condition assessments, decision-making, repairs and maintenance, and future priorities into its strategic model. Like many asset management frameworks, most of the components in the framework support performance management. Alternatively, the holistic asset management framework incorporates sustainability and resilience management concepts into the asset management discussion (Gay and Sinha, 2013). Sustainability and resilience management are needed for an asset management framework to be truly holistic. Performance focuses on the physical functioning of assets as they pertain to providing the desired Level of Service (LoS). Sustainability and resilience management concepts bring goals and standards to a utility that include the well-being of the community and the environment as well as preparation for disaster events to support the desired level of service based on performance, as shown in Figure 13.

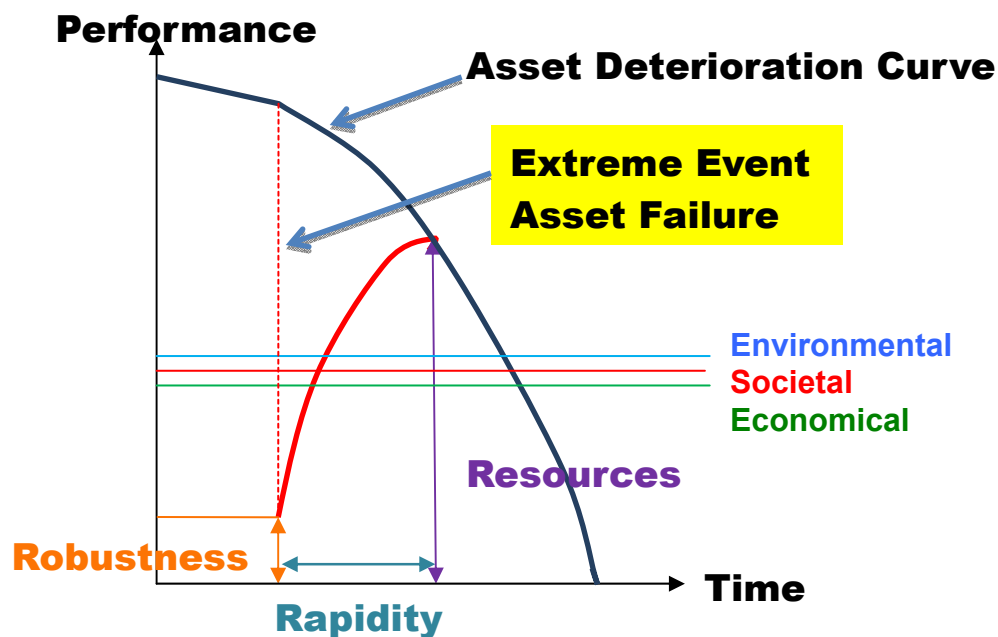


Figure 13. Infrastructure Management based on Sustainability and Resiliency

6. Development of Centralized Water Pipeline Infrastructure Data Center

The data will be collected from various utilities geographically distributed across the US. Throughout the project the research team will solicit additional participating utilities of all sizes and from all regions. Virginia Tech (VT) has already signed a Memorandum of Understanding (MoU) with various participating utilities for water infrastructure related data transfer. These data will be uploaded to the standard data model template and will then be transferred to the VT GIS Server infrastructure for integration with the Web-based component of PIPEiD described above. The Enterprise GIS unit of the Information Technology organization at VT centrally manages VT's GIS server infrastructure. Many of the problems of data transmission from utilities, central storage across distributed nodes, and the integration with Web-based mapping interfaces have already been resolved. The VT-FTP server has been provisioned to receive data from a large number of utilities, and an automated account provisioning system was developed for this purpose to facilitate ease of access by utilities while assuring security of sensitive data. The research team will develop translation and mapping algorithms to enable legacy databases, which do not comply with the data standards, to interface with PIPEiD. To simplify data aggregation, the research team will provide methodologies and protocols for baseline, O&M, and forensic data reporting and mapping. Multiple pilot studies will be conducted with various utilities to verify these protocols. There are several existing geo-databases relevant to water infrastructure that are available online or published. These geo-databases will help the research team in comparing the data provided from the utilities as well as filling in missing data parameters. Such ancillary datasets will be integrated into both the spatial data warehouse and the interactive Web application components of PIPEiD for ease of access, thereby enabling the environmental conditions to be directly visualized within the same interface as the infrastructure data. This will also enable development of models and tools that depend on these ancillary variables as inputs for advanced water pipeline management (Figure 14).

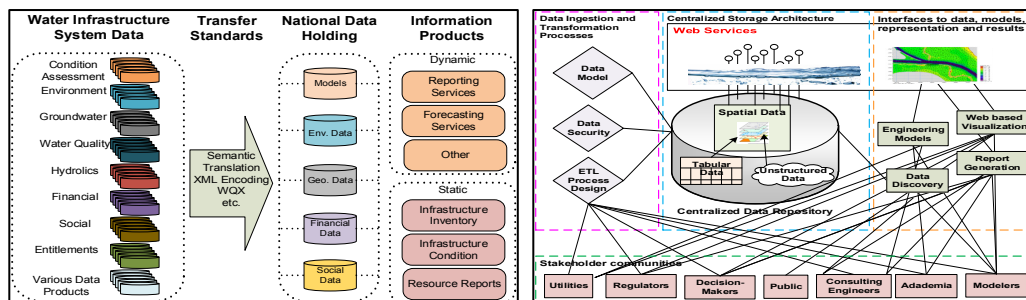


Figure 14. Standardized Data (left) and Modeling Platform (Right) for Analysis

7. Model Development, Demonstration, Benchmarking, and Validation

Model capabilities are limited not by mathematics, but by a lack of consistent and sufficient data and by the complexity of analysis. Emphasis has been on modeling rather than data integration, because modeling is easier and less costly than data aggregation and integration; however, without access to integrated physical, operational, environmental, social, and financial data, models are of limited use. PIPEiD will serve as the nation's central depository for standardized data, models,

and tools for advanced asset management of water infrastructure systems. PIPEiD will be an interoperable platform for the development, implementation, and benchmarking of models, to enable accurate quantitative analysis of performance, reliability, and sustainability of water infrastructure systems. The problem with most models and tools is a lack of completeness in the dataset, which leads to an inability to achieve reliable, system level evaluation and validation. In fact, there is currently no ability to calibrate, benchmark, verify, and visualize the reliability of models for use by any utility. PIPEiD will perpetually enable researchers to benchmark and validate the reliability and accuracy of models and tools utilizing the PIPEiD dataset. All models and tools developed for PIPEiD will be freely available to water industry. Valuable numeric knowledge is hidden in the huge amount of drinking water and wastewater pipeline infrastructure related data collected from various sources. It is important to “mine” useful patterns, rules, and anomalies from the historical data and utilizes that knowledge to help pipeline managers make effective decisions. The data analysis and mining modules will perform two major tasks: (1) identify frequent patterns within key factors of pipe performance and (2) discover abnormal events. The PIPEiD data analysis techniques consist of the following: (1) a set of data mining techniques that can identify key factors influencing the performance of the pipelines; and (2) a self-evolving knowledge base which integrates the experiences and knowledge of engineers to address the limitations of models and tools as well as to make better decisions regarding subsequent condition assessment and rehabilitation activities. The research team will benchmark and verify data analysis techniques for reliable use. The research team will address the advanced data visualization issue by designing an online data processing system based on standards for Web services and spatial data management, i.e., GML (Geographical Markup Language) and SVG (Scalable Vector Graphics). First, Web service protocols [ex: SOAP (Simple Object Access Protocol), UDDI (Universal Description, and Discovery and Integration)] will be utilized to facilitate the communications among various data sources on the Internet. Through standard XML-based Web services, water infrastructure data can be intelligently published and subscribed for dynamic information retrieval. Second, our proposed system will provide data analysis and data mining Web services to disseminate information and support input and query from utility with real-time updates. Third, formulating data in XML and delivering the contents in SVG will provide adaptive dissemination and visualization provided for the utility professional with various access devices. We will also develop Application Program Interface (API) and broadcast technology for mobile devices (Figure 15).

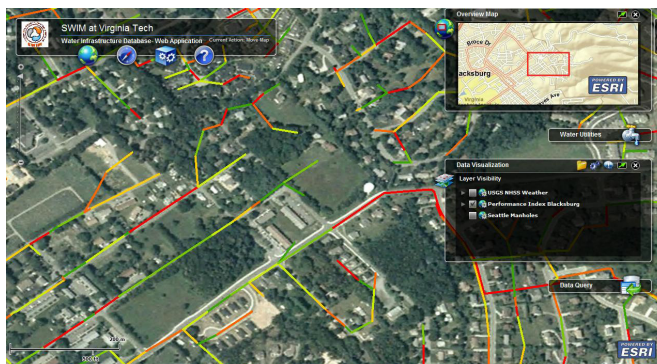


Figure 15. Water Pipeline Infrastructure System Data Visualization for Decision Support

8. Conclusions and Recommendations

The objectives of the proposed project are to provide a unified platform for the nation's water pipeline infrastructure data and information, to make it universally accessible and useful, and to provide access to the data sources, tools, and models that enable the analysis, simulation, visualization, and evaluation of the behavior of water infrastructure systems. The proposed research plan represents a synthesis of diverse types of data, models, analysis, and expertise for improving our understanding, forecasting, and management of water infrastructure systems. Project will provide capabilities to identify, understand, model, simulate, analyze, and visualize the dynamics of sustainable water infrastructure systems. PIPEiD project will explore model research across broad natural, socio-economic, and built domains to yield an integrated water system management model for predictive understanding by leveraging existing data, models, and simulation tools. The project has five critical and integrated core components:

- A national water pipeline data standards and system architecture for database
- An open-source platform to support integrated pipeline infrastructure modeling
- Novel research that engages big-data, models, modelers, users, and stakeholders
- Community Outreach that fosters and trains a community of stakeholders
- Maintenance and development help for freely available pipe infrastructure models

The proposed research will result in significant advancement to understanding and managing water pipeline infrastructure systems by utilizing spatio-temporal data through modeling, simulation, and visualization for sustainable and resilient water infrastructure systems. The stakeholders within water infrastructure community must work together to accelerate progress toward these highly multi-disciplinary research, education, training, and outreach needs.

9. References

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